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# Higher-order Laplacian renormalization

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The renormalization group is a pillar of the theory of scaling, scale invariance and universality in physics. Recently, this tool has been adapted to complex networks with pairwise interactions through a scheme based on diffusion dynamics. However, as the importance of polyadic interactions in complex systems becomes more evident, there is a pressing need to extend the renormalization group methods to higher-order networks. Here we fill this gap and propose a Laplacian renormalization group scheme for arbitrary higher-order networks. At the heart of our approach is the introduction of cross-order Laplacians, which generalize existing higher-order Laplacians by allowing the description of diffusion processes that can happen on hyperedges of any order via hyperedges of any other order. This approach enables us to probe higher-order structures, define scale invariance at various orders and propose a coarse-graining scheme. We validate our approach on controlled synthetic higher-order systems and then use it to detect the presence of order-specific scale-invariant profiles of real-world complex systems from multiple domains.

The renormalization group (RG)[1] is a cornerstone of modern theoretical physics as it allows us to study how a physical system depends on the scale of observation, defining universality classes and, importantly, formalizing the concept of scale invariance. Although the RG has been a powerful tool for understanding a broad class of physical systems, extending its framework to complex networks has posed a recent and substantial challenge, mainly due to the correlations between scales caused by small-world effects[2]. This challenge has gained substantial attention[3–10] due to its potential to provide insights into the multiscale structural organization of complex networks.

Notable approaches[6,11] are based on the hypothesis that the network's nodes can be naturally embedded in a metric space, whose geometry provides a natural way to identify groups of nearby nodes, reminiscent of spin blocks in the traditional real-space RG process[12]. However, this perspective encounters a fundamental limitation: networks are inherently topological structures—devoid of geometry—and, thus, need a topological notion of RG[7,13].

Diffusion provides such a notion: it is a dynamical process that one can define on any combinatorial structure and depends only on its topology. In particular, diffusion on graphs describes the dynamics of information flowing from node to node through edges, eventually becoming uniformly distributed on its connected components. This process is formalized as a first-order system of linear differential equations specified by the graph Laplacian matrix $\mathbf{L}$. One can then see the diffusion time as a resolution parameter: at short times, information only diffuses to the neighbouring nodes, revealing the local structure; at longer times, diffusion reaches nodes further apart, revealing the global structure. A recent proposal—the Laplacian renormalization group (LRG)[10]—leverages this observation to produce coarser descriptions of a network's structure by identifying groups of nodes that are strongly linked by diffusion at a given scale. Moreover, one can adopt this same approach to define an informational notion of scale invariance, based on the properties of the diffusion process via the Laplacian spectrum[14], and thus different from the canonical concept of scale freeness, which instead depends only on the degree distribution. Both these results hinge on the formalism of network density matrices[15] to describe the complete behaviour of information diffusion at a given scale.

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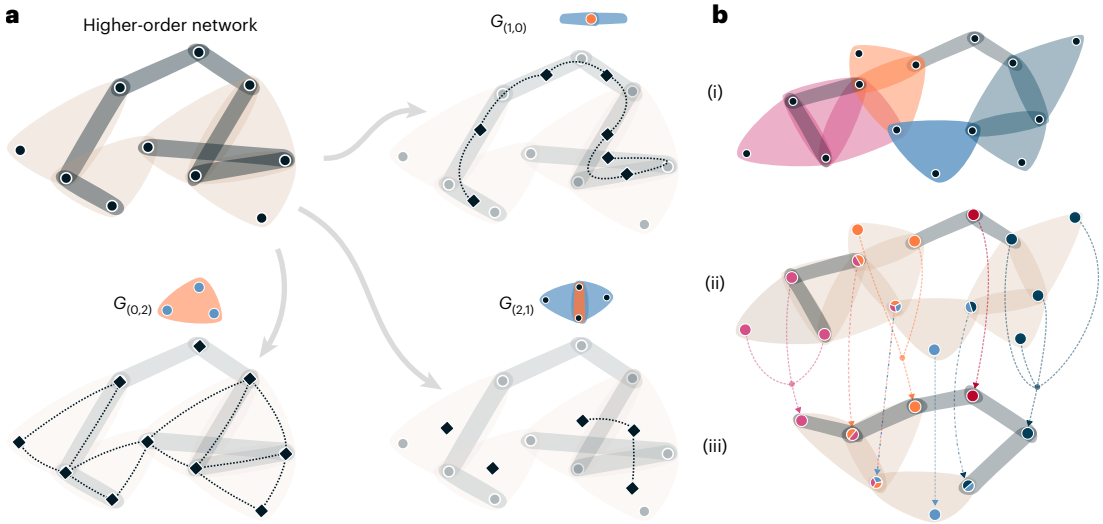



**Fig. 1 | Cross-order Laplacian renormalization scheme: partition and coarse graining. a**, Schematic of how the adjacency graphs are built for a hypergraph. Notice how there can be cases like $G_{(0,2)}$, in which the edges are weighted (here represented with multiedges). **b**, Pictorial representation of our higher-order coarse-graining scheme with $k = 2$: (i) a partition of the two hyperedges, represented here with colour, is obtained from $\boldsymbol{\rho}(\tau)$; (ii) each vertex inherits a signature containing the labels of the two hyperedges it belongs to; (iii) vertices with the same signature are glued together and the hyperedges are induced from the starting hypergraph.

Networks, however, are only part of the story. Great attention has recently been devoted to the study of higher-order networks: networks that encode multinode interactions, going beyond the pairwise interactions of traditional networks[16–18]. Higher-order interactions are present in many natural systems and drastically impact most dynamical processes, such as random walks[19–21], diffusion[22,23], spreading[24–30], coordination[31–37] and synchronization[38–43].

However, little work still exists on RG approaches to higher-order networks. A direct generalization of the Laplacian RG approach[10], based on the multiorder Laplacian[39], was recently proposed[44]. This proposal is, however, node centric: it only considers the diffusion of information from node to node. A parallel research line[14,45,46] focused on specific families of simplicial complexes and used RG techniques to compute some notable statistical properties.

Here we propose a general RG scheme for arbitrary higher-order networks. Our approach uses a higher-order notion of diffusion that we formally define by introducing the cross-order Laplacian. In this new diffusion process, information can flow through hyperedges of any order $k$ via hyperedges of any other order $m$. This proposal provides a natural generalization of previous ones[44,47], which are restricted to node–node diffusion. By studying the properties of this diffusion via the cross-order Laplacians, our approach allows us to probe the existence of characteristic scales, or—crucially—their absence (scale invariance) in higher-order networks at each order. In particular, we first define the appropriate Laplacian matrices to describe generic higher-order diffusion. We then leverage them to define (1) a higher-order notion of informational scale invariance through von Neumann entropy and entropic susceptibility and (2) an explicit RG scheme informed by a chosen higher-order diffusion process. Using these tools, we extract a cross-order scale signature in higher-order networks obtained from synthetic models and real-world data and show that in most cases, scale invariance is found only under the lens of specific orders, suggesting the existence of underlying order-specific processes.

## Cross-order Laplacians

Let $\Delta$ be a higher-order network (or hypergraph) on a finite set of vertices (or nodes) $V$, that is, a family of subsets of $V$ named hyperedges. We want to find whether the hypergraph possesses characteristic scales or is scale invariant from the perspective of $k$-node hyperedges. Given that a diffusion process has been shown to be akin to a telescopic scanner of a (pairwise) network, capable of extracting multiscale information about its structure[10], we wish to extend the same idea to higher-order networks.

A diffusion-like process on the $k$-hyperedges of a higher-order network can be defined in multiple ways, corresponding to multiple Laplacian matrices. In the cases where $\Delta$ is a simplicial complex, that is, every subset of a hyperedge (simplex) is a hyperedge, a diffusion process of this form[22] can be defined with the combinatorial Hodge Laplacian[48], making it a natural candidate for our approach. However, diffusion through the Hodge Laplacian does not correspond to a 'standard' diffusion process as, for instance, the total amount of information flowing between hyperedges is not conserved (Supplementary Section 1). Due to this lack of clear physical interpretability, it is difficult to directly use it in the Laplacian renormalization framework proposed in another work[47].

Thus, we define a new family of Laplacian matrices that can describe a plethora of higher-order relations and maintaining a form analogous to the canonical graph Laplacian. We do this by taking inspiration from the general theory of combinatorial complexes[49] and the hypergraph Laplacian[50]. Fixing $k \in N$ as the diffusion order and $m \in N$ as the interaction order, we define the adjacency graph $G_{(k,m)}$ associated to the hypergraph as a weighted graph whose nodes are the $k$-hyperedges and the edges are their adjacency relations given by $m$-hyperedges (Fig. 1a). In detail, two $k$-hyperedges $h_1$ and $h_2$ are $m$-adjacent if there exists an $m$-hyperedge $h_3$ such that $h_3 \subseteq h_1 \cap h_2$, if $m < k$, as well as $h_3 \supseteq h_1 \cup h_2$, if $m > k$. In particular, $G_{(0,1)}$ is the graph underlying the higher-order network and $G_{(1,0)}$ corresponds to its line graph[51]. To each (weighted) adjacency graph, we can associate its (weighted) Laplacian, which we call a cross-order Laplacian, through the usual formula:

$$\mathbf{L}^{\times}_{(k,m)} = \mathrm{diag}(\deg_{(k,m)}) - \mathbf{A}_{(k,m)}, \tag{1}$$

where $\mathbf{A}_{(k,m)}$ is the (weighted) adjacency matrix associated to the adjacency graph and $\deg_{(k,m)}$ is the vector containing the higher-order $(k,m)$-degrees (Methods).

## Measuring scale invariance

From now on, we will consider the high-order diffusion process on $k$-hyperedges through $m$-hyperedges on a given hypergraph $\Delta$

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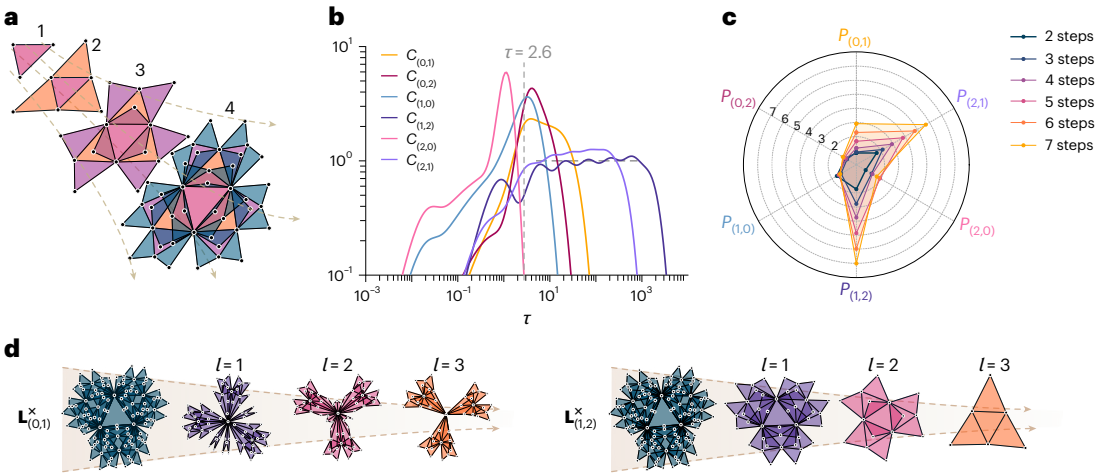



**Fig. 2 | Scale invariance and renormalization in pseudofractal simplicial complexes. a**, Graphical depiction of the first three steps of constructing the pseudofractal simplicial complex with two dimensions. **b**, Entropic susceptibilities of all the cross-order Laplacian matrices, computed for the two-dimensional pseudofractal simplicial complex built with six steps (1,095 vertices). **c**, SIPs varying with the number of steps with which the simplicial complex is built. **d**, Left: the first three steps of the $\mathbf{L}^{\times}_{(0,1)}$ renormalization scheme with $\tau^* = 0.2$. Right: the first three steps of the $\mathbf{L}^{\times}_{(1,2)}$ renormalization with $\tau^* = 2.6$.

and, thus, omit the $(k, m)$ notation. The diffusion properties of the higher-order network at time $\tau$ are completely described by the density matrix

$$\hat{\boldsymbol{\rho}}(\tau) = \frac{e^{-\tau \mathbf{L}^{\times}}}{\mathrm{Tr}\left(e^{-\tau \mathbf{L}^{\times}}\right)}, \tag{2}$$

its associated von Neumann entropy

$$S(\tau) = -\mathrm{Tr}\left(\hat{\boldsymbol{\rho}} \log[\hat{\boldsymbol{\rho}}]\right) \tag{3}$$

and entropic susceptibility[47] (Methods)

$$C(\tau) = -\frac{\mathrm{d}S(\tau)}{\mathrm{d}\log[\tau]}. \tag{4}$$

These quantities allow us to directly apply the cross-order Laplacian to investigate the hypergraph's $k$th-order properties at different scales. The $k = 0$ case has been fruitfully explored in the literature to extract key information about the network's structural organization[15,52–54].

In particular, the maxima and minima of the entropic susceptibility, associated with times of fast deceleration and acceleration of the diffusion process, were shown to correspond to the presence of characteristic scales[10,47]. Most importantly, when $C(\tau)$ is constant over a time range $I = [\tau_{\min}, \tau_{\max}]$, we say that $\Delta$ exhibits informational scale invariance[10] in $I$. For instance, it is known that the (network, $k = 0$ and $m = 1$) entropic susceptibility $C_{(0,1)}(\tau)$ shows a large plateau in the case of grid graphs, Barabási–Albert networks and random trees, all of which are examples of self-similar structures. Interestingly, entropic susceptibility and the associated notion of scale invariance can be related to the concept of spectral dimension[55–57], which intuitively measures the dimensionality 'perceived' by a diffusion process taking place on a manifold or, in our case, a graph (Supplementary Section 2).

To quantify whether a higher-order network exhibits scale invariance at order $k$ via order $m$, we define the scale-invariance parameter (SIP) $P_{(k,m)}(\epsilon)$, as the logarithmic lifespan of the longest connected plateau of $C_{(k,m)}(\tau)$ with respect to a given tolerance $\epsilon > 0$ on its 'flatness' (Methods). If $P_{(k,m)}(\epsilon)$ is large, we say that the hypergraph is scale invariant at order $k$ via order $m$, whereas if it is close to 0, then $C_{(k,m)}(\tau)$ is peaked, thereby signalling the presence of characteristic scales.

## Cross-order Laplacian renormalization scheme

Let us assume that a hypergraph $\Delta$ has been recognized as scale invariant with respect to $C_{(k,m)}(\tau)$ (equation (14)). We would now want an algorithmic method to reduce it to a smaller, equivalent one that can still be recognized as scale invariant. Similar to the case of networks, we devise a higher-order Laplacian renormalization scheme consisting of the following steps:

1. first, compute the cross-order Laplacian matrix $\mathbf{L}^{\times}_{(k,m)}$;
2. choose a diffusion time $\tau^* > 0$ corresponding to the scale at which to 'zoom out';
3. compute a partition of the $k$-hyperedges from the elements of the density matrix $\boldsymbol{\rho}_{(k,m)}(\tau^*) \propto \exp(-\tau^* \mathbf{L}^{\times}_{(k,m)})$, such that hyperedges in the same set are strongly linked by the diffusion process at time $\tau^*$;
4. coarse grain $\Delta$ by merging its vertices according to the partition to obtain a new, smaller $\Delta'$.

The process is then repeated, resulting in a sequence of hypergraphs with a decreasing or constant number of vertices (Methods and Supplementary Section 3). We name such a sequence as a renormalization flow.

## Higher-order scale invariance

We now show explicit examples of the applications of this framework. We first focus on synthetic models of simplicial complexes to confirm that in controlled situations, the cross-order RG exactly recovers the scale-invariant structure and order of the underlying system. After this confirmation, we extract the cross-order scale signatures of some real-world datasets.

### Pseudofractal simplicial complexes

The family of scale-free pseudofractal simplicial complexes[58] provides a simple example of self-similar higher-order networks. Simplicial complexes in this family are built starting with a single $k$-simplex and by iteratively attaching a $k$-simplex to each $(k-1)$-simplex already present in the complex (Fig. 2a).

We expect the evident hierarchical nature of these objects to be visible in the entropic susceptibility curve of one or more diffusion processes defined on it.

Figure 2b illustrates how the different entropic susceptibilities $C_{(k,m)}$ display different non-trivial behaviours. Most importantly, the

curves associated with $\mathbf{L}^{\times}_{(1,2)}$ and $\mathbf{L}^{\times}_{(2,1)}$ show oscillating plateaus that span multiple orders of magnitude in $\tau$. These correspond to the different, well-separated scales resulting from the iterative construction process, which can be interpreted as an approximate form of higher-order informational scale invariance. Surprisingly, even if the network underlying the simplicial complex is scale free[59], the curve of $\mathbf{L}^{\times}_{(0,1)}$ (the canonical graph Laplacian) is peaked, meaning that the self-similarity of the structure is not visible at the vertex-edge level.

These results are consistently observed when we increase the number of steps with which the simplicial complex is built. We find that almost all the SIPs $P_{(k,m)}$ increase with the number of construction iterations (Fig. 2c), with the highest values being consistently obtained by $P_{(1,2)}$ and $P_{(2,1)}$, which correspond to the cross-order Laplacians naturally associated with the complex's growth (addition of triangles along the edges). Moreover, the $P_{(2,0)}$, $P_{(1,0)}$ and $P_{(0,2)}$ values remain approximately constant, whereas $P_{(0,1)}$ slowly increases, indicating the presence of a growing plateau (it is invisible in Fig. 2b; we show the realization results with six steps). This observed behaviour is also consistent with previous results[14] on the spectrum of $\mathbf{L}^{\times}_{(0,1)}$ of pseudofractal simplicial complexes, which shows a power-law behaviour for small eigenvalues when the number of vertices is large, that is, a plateau in the entropic susceptibility[47]. However, the scale-invariant behaviour is much more easily and quickly detectable when considering the entropic susceptibility of Laplacians associated with the complex's intrinsic growth process (between edges and triangles), even when the number of vertices is still small.

We confirm this observation by applying the renormalization scheme to a pseudofractal simplicial complex with two dimensions and visualizing its evolution at each step. The renormalization based on $\mathbf{L}^{\times}_{(1,2)}$ preserves the structure of the simplicial complex, perfectly reversing its construction process (Fig. 2d, right). By contrast, the renormalization flow based on $\mathbf{L}^{\times}_{(0,1)}$ destroys the pseudofractal structure and rapidly collapses the central vertices, resulting in a 'star-shaped' simplicial complex with no apparent relation with the original one (Fig. 2d, left). In Supplementary Section 5, we thoroughly explore the renormalization of the pseudofractal simplicial complex under different Laplacians and $\tau^*$ values.

## Network geometry with flavour

Although the higher-order scale invariance for pseudofractal simplicial complexes is a relevant consistency check, this result is not particularly striking due to their evident hierarchical nature. We show here that analogous results can be found in the case of heterogeneous simplicial complexes, similar to what was previously observed in random scale-free networks and random trees[10]. To this end, a natural candidate is the family of simplicial complexes given by the network geometry with flavour (NGF) model[59,60]. Indeed, the NGF model in dimension $d$ can generate both hyperbolic manifolds and scale-free growing simplicial complexes by progressively attaching $d$-simplices to $(d-1)$-simplices in a stochastic manner (Supplementary Section 6A provides details on the model). Given their scale-free nature at all levels, our focus is on NGF simplicial complexes with flavour $s=1$.

To manage computational complexity, we calculate curves only for relations predicted to be scale free, alongside the vertex-edge (0,1) relation for comparison.

Although the NGF construction process is similar to the pseudofractals, the randomness component does not allow for a clear separation of the hierarchical scales. The averaged curves of $C_{(d,d-1)}$ exhibit a smooth plateau anticipated by a small peak, which grows with higher dimensions, indicating the presence of a distinct microscopic scale. This is because the $(d-1,d)$-adjacency graph (of both $d$-dimensional NGFs and pseudofractals) is composed by $(d+1)$ cliques arranged in a tree-like structure, as $d$-simplices have $d+1$ $(d-1)$-faces. Thus, the first peak corresponds to the microscale associated with these cliques (Supplementary Section 4 provides a visualization). Afterwards, a scale-invariant regime ensues, illustrated by the near-perfect plateau of $C_{(d-1,d)}$. However, despite being associated with power-law degree distributions, other curves lack plateaus, showcasing non-trivial structural organization at specific scales.

In Fig. 3b, a renormalization based on a higher-order relation ($\mathbf{L}^{\times}_{(1,2)}$) shows better preservation of the structure of a two-dimensional NGF simplicial complex with respect to the standard vertex-edge Laplacian renormalization. In particular, the $\mathbf{L}^{\times}_{(0,1)}$ renormalization drastically reduces the number of vertices, collapsing them into a single super-vertex at the first peak of $C_{(0,1)}$ (Fig. 3c, left). Conversely, $\mathbf{L}^{\times}_{(1,2)}$ renormalization compresses the simplicial complex more gradually, revealing a clear transition point just before the first peak (Fig. 3c, right). This distinction is further emphasized by tracking the evolution of the entropic susceptibility after one step of the renormalization flow (Fig. 3d), where the plateau of $C_{(1,2)}$ is preserved by $\mathbf{L}^{\times}_{(1,2)}$ but destroyed by $\mathbf{L}^{\times}_{(0,1)}$.

## Higher-order scale invariance in real data

We can leverage SIPs to provide effective higher-order signatures describing their hierarchical nature in a multifaceted way. Note that in general, unlike the regular cases discussed above, real networks are not the result of a simple construction process, and thus, there is no clear way to identify a priori the pair (or pairs) $(k,m)$ that can exhibit scale invariance. Moreover, in a real-world context, we cannot expect the presence of scale invariance to be as clear-cut, with well-defined plateaus as in the synthetic cases. In turn, fixing a maximum order $k_{\max}$, the most complete view is obtained by looking at all the $k_{\max}(k_{\max}-1)$ entropic susceptibility curves up to that order.

We take real higher-order network datasets (Supplementary Section 7D) and consider their interactions up to the fourth order, that is, hyperedges with a maximum of five nodes. For each hypergraph, we compute all the $C_{(k,m)}$ values with the associated $P_{(k,m)}$ values and build a null model by doing the same on their adjacency graphs randomized with a degree-preserving configuration model. In this way, we are able to obtain an effective signature of the hypergraphs encoding the amount of scale invariance in each of its higher-order relations (Fig. 4a).

Different behaviours emerge: the 'flower-like' shape for the high-school contacts[61] hypergraph, the 'fan-like' shape—with higher values of scale invariance in some specific orders—of the Senate bills[62], House bills[62], the Diseasome[63] and the National Drug Code substances[64] hypergraphs. Moreover, as for the Enron email dataset[64,65], there can be a more uniform distribution of the SIP values, without a clear dominance of orders. Note that when the $k$-hyperedges are too sparse, information cannot flow through the $m$ hyperedges, resulting in zero-valued SIPs for those specific orders.

Interestingly, even when we do not consider genuine higher-order networks, but rather focus on networks, it is still possible to obtain meaningful information through our approach. We consider real pairwise network datasets (Supplementary Section 7C) and compute their associated clique complexes up to second order, that is, build simplicial complexes by considering their cliques of three nodes as two hyperedges. Notice that in this way, we artificially lift a (pairwise) network to a higher-order one, and consequently, any resulting higher-order properties should be interpreted as describing the structural organization of cliques of nodes and not of 'genuine' higher-order interactions.

In Fig. 4c, we show the SIPs associated to some of these clique complexes, together with the null model. We find a variety of different behaviours, like the E-road network[66] and the network science co-authorship network[67], which are strongly scale invariant in the standard sense, that is, in the relation between vertices and edges (high $P_{(0,1)}$ and $P_{(1,0)}$ values). Despite the inherent pairwise nature of the networks, we find a marked presence of high-order scale invariance in some of them, like the University Rovira i Virgili Email network[68], the *C. elegans* metabolic network[69–71] and the fictional A Song of Ice and

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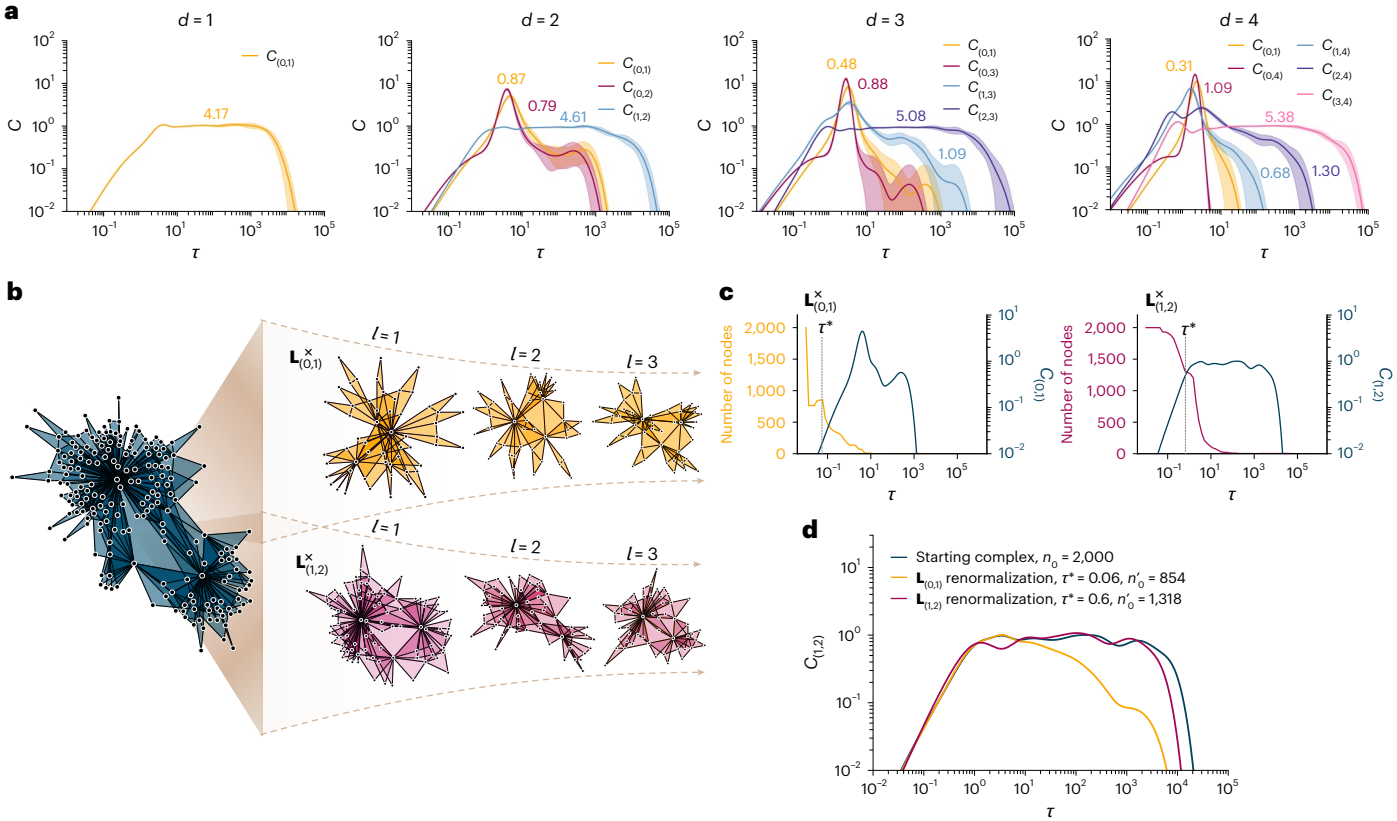



**Fig. 3 | Higher-order Laplacian renormalization scheme applied to NGF simplicial complexes. a**, Entropic susceptibilities, together with their 95% confidence intervals over ten repetitions, of the NGF simplicial complexes of flavour $s = 1$, dimensions $d \in \{1, 2, 3, 4\}$, $\beta = 5$ and 3,000 vertices. The numbers display the SIPs. **b**, A small two-dimensional NGF simplicial complex is renormalized using $\mathbf{L}^{\times}_{(0,1)}$ (top) and $\mathbf{L}^{\times}_{(1,2)}$ (bottom). **c**, Left: $C_{(0,1)}$ together with the number of vertices of the complex after one step of $\mathbf{L}^{\times}_{(0,1)}$ renormalization as a function of $\tau$. Right: $C_{(1,2)}$ together with the number of vertices of the complex after one step of $\mathbf{L}^{\times}_{(1,2)}$ renormalization as a function of $\tau$. **d**, Evolution of the entropic susceptibility $C_{(1,2)}$ after the first step of the two types of renormalization considered. The starting NGF simplicial complex has 2,000 vertices and is reduced to 854 vertices by $\mathbf{L}^{\times}_{(0,1)}$ ($\tau^* = 0.06$) and to 1,318 by $\mathbf{L}^{\times}_{(1,2)}$ ($\tau^* = 0.6$).

Fire social network[72]. We stress again that this higher-order description is obtained without any assumptions on the presence of genuine higher-order interactions in these systems, but it is rather based only on nodes, edges and the cliques they form. However, some aspects of the hierarchical scale-invariant structure of a network may be hidden from the node-centric point of view, but can be evident when looking at how simplices (in this case, cliques) are related to one another. This is in line with similar results on the persistent homology of weighted networks[73].

Finally, we use SIPs as coordinates to embed the datasets in $\mathbb{R}^2$ (Supplementary Section 7C,D). In Fig. 4b,d, the linear projection—linear discriminant analysis[74]—of the resulting point cloud shows heterogeneity in the amount of higher-order scale invariance present in each dataset. Surprisingly, despite the coarseness of this measure, we see that the information contained in the SIPs is enough to cluster the networks according to their nature, suggesting the presence of specific higher-order signatures associated with each data type.

## Discussion

By leveraging the LRG, we developed a method to investigate the structure of higher-order networks via a renormalization procedure based on the connectivity structure of higher-order relations, as encoded in the proposed cross-order Laplacian. We showed that the renormalization scheme can exactly revert the construction of higher-order structures that are self-similar by construction (the pseudofractal) and detect the correct order of the dominating growth mechanism in scale-invariant complexes with induced heterogeneous lower-order structures (the NGF model). Armed with these results on controlled synthetic systems, we used the entropic susceptibilities obtained in our scheme to build scale-invariant profiles for a set of real-world systems, revealing both different unexpected kinds of characteristic scale and scale invariance at various orders and commonality of such profiles across systems belonging to the same domain.

From a technical point of view, the cross-order Laplacians bear both similarities and differences with previously defined higher-order Laplacians, particularly the Hodge (or combinatorial) Laplacian[22] $\mathcal{L}_k$ and the multiorder Laplacian $\mathbf{L}^{(\mathrm{mul})}$ (ref. 39). In fact, both $k$-order Hodge Laplacian and cross-order Laplacians $\mathbf{L}^{\times}_{(k,m)}$ (for any $m$) are defined on the hyperedges of order $k$ and, thus, are $n_k \times n_k$ matrices, where $n_k$ is the number of $k$-hyperedges. However, the Hodge Laplacian is limited to adjacencies defined by boundary and co-boundary relations and, thus, with $m = k \pm 1$, whereas the cross-order can capture arbitrary adjacencies through any $m$-hyperedges. Conversely, the multiorder Laplacian $\mathbf{L}^{(\mathrm{mul})}$ is defined as the weighted sum of Laplacians defined on nodes and adjacent via hyperedges of any $m$ order. It can be rewritten in our notation as $\mathbf{L}^{(\mathrm{mul})} = \sum_{m=1}^{M} \omega_m \mathbf{L}^{\times}_{(0,m)}$, where $\omega_m$ is an arbitrary weighting scheme and $M$ is the maximum order considered. Similarly, although in this work we focused on renormalization based on specific pairs $(k, m)$, we note that important future work should understand the effect of extending the cross-order renormalization scheme to a multicross-order scheme, which could be achieved by considering combinations of hyperedges of dimensions higher or lower

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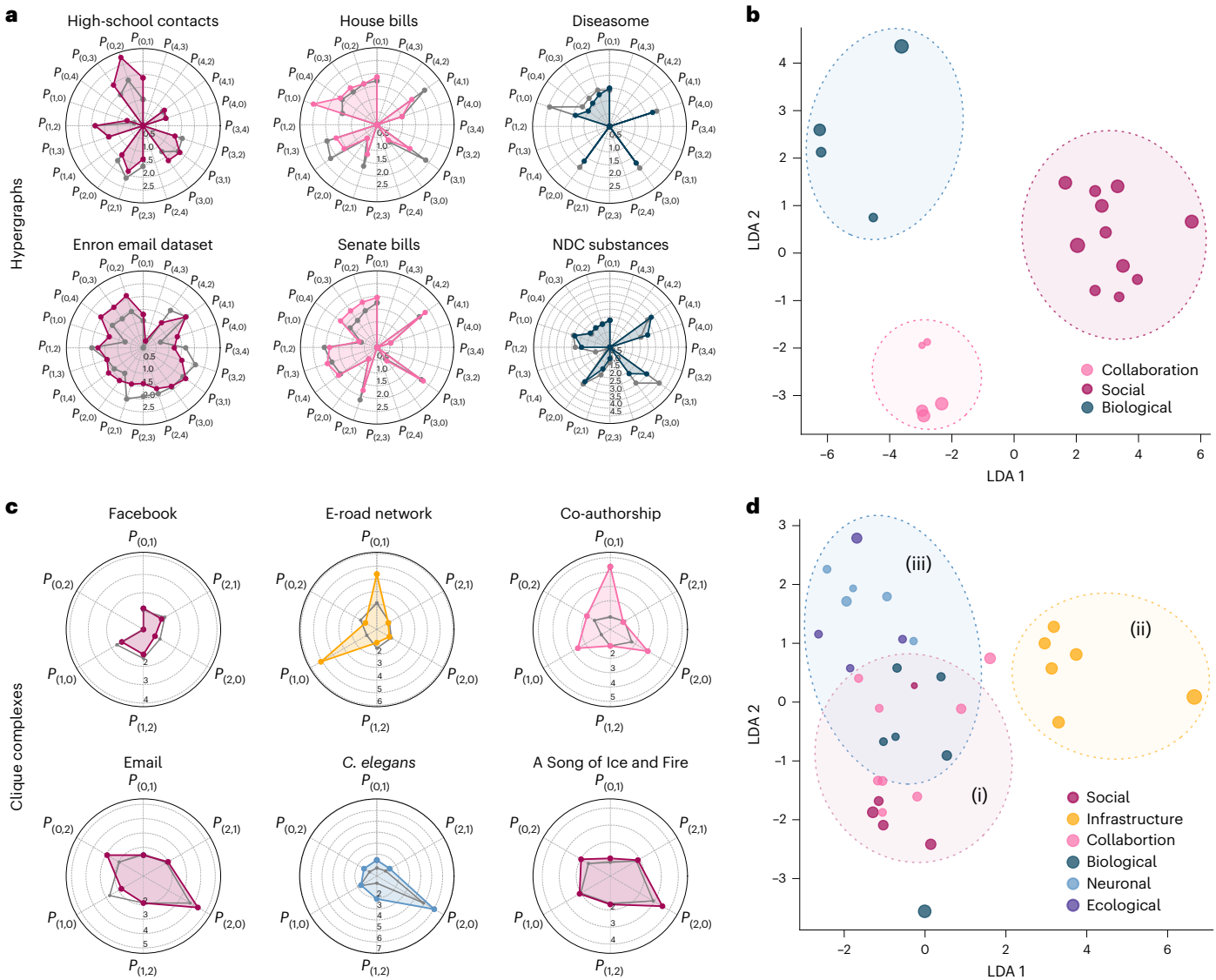



**Fig. 4 | Higher-order scale invariance in real data. a**, SIPs $P_{(k,m)}$ of six hypergraphs—truncated at order 4—obtained from real-world datasets. The grey lines and the associated regions represent the mean and 95% confidence interval for the $P_{(k,m)}$ values of the null model over ten repetitions. NDC, National Drug Code. **b**, Linear discriminant analysis (LDA) projection of a larger set of 20 real hypergraphs of different types, denoted by different colours, and with size proportional to the value of their highest SIP. The coloured regions highlight the clusters of nearby points that belong to the same class of hypergraphs. **c**, SIPs $P_{(k,m)}$ of six second-order clique complexes obtained from real-world network datasets. The grey lines and the associated regions represent the mean and 95% confidence interval for the $P_{(k,m)}$ values of the null model over ten repetitions. **d**, LDA projection of 34 real clique complexes of different types, denoted by different colours, and with size proportional to the value of the network's highest SIP. The coloured regions highlight the clusters of nearby points that belong to the same class: social origin (i), infrastructural (ii) and biological origin (iii).

than a certain threshold on the interaction order (for example, $\mathbf{L}_k^{(\mathrm{mul}\times)} = \sum_{m\in\{\mathbf{m}\}} \omega_m \mathbf{L}^{\times}_{(k,m)}$ over a set $\{\mathbf{m}\}$). Also, although this work mainly focused on simplicial complexes for convenience and clearness of exposition, any combinatorial structure[49] built with a set of ranked sets is amenable to our scheme.

Finally, our results provide a new lens to address questions on the origin of different higher-order invariant structures in various domains[75,76], on their effects on dynamical processes taking place on them[25,28,28] and their reducibility[77], as well as to the limits they might pose on the predictability and reconstruction of complex systems[78].

## Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41567-025-02784-1.

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# Methods

## Higher-order networks and their structure

**Higher-order networks.** Let $\Delta$ be a higher-order network (also named a hypergraph) on a finite set of vertices $V$, that is, a family of subsets of $V$. Any element of $\Delta$ is called a hyperedge and its order is defined as its cardinality reduced by one. Thus, a vertex will be a 0-hyperedges; an edge, a 1-hyperedge; a triangle, a 2-hyperedges; and so on. If $\eta \in \Delta$ is a subset of a $k$-hyperedge $\sigma$, then we say that $\eta$ is a face of $\sigma$. We write $\Delta_k$ to denote the set of $k$-hyperedges of $\Delta$ and call $n_k$ its cardinality. The object $\Delta$ is a simplicial complex when closed under inclusion, that is, $\sigma \in \Delta, \eta \subseteq \sigma \Rightarrow \eta \in \Delta$. In such a case, the hyperedges are called simplices.

Having fixed an order $k \in N$, we want to find whether the hypergraph possesses characteristic scales or is scale invariant from the perspective of $k$-hyperedges. In ref. 10, it is argued that a diffusion process can be seen as a telescopic scanner of a (pairwise) network capable of extracting multiscale information about its structure. This fact suggests studying our hypergraph's $k$th-order properties through a diffusion process on the $k$-hyperedges.

**Cross-order Laplacians.** A diffusion-like process on the hyperedges of a higher-order network can be defined in multiple ways, each associated with a different Laplacian matrix. In the case of simplicial complexes, the $k$th-order combinatorial Hodge Laplacian[48] is the most commonly considered[79,80], mainly for its deep connections with topology[79]. A diffusion process on the $k$-hyperedges[22] can indeed be defined with the Hodge Laplacian, making it a natural candidate for our approach. However, it is possible to see that diffusion through the Hodge Laplacian does not correspond to a 'standard' diffusion process as, for instance, the total amount of information flowing between the hyperedges is not conserved (Supplementary Section 1). Due to this lack of clear physical interpretability, it is difficult to directly use it in the Laplacian renormalization framework proposed in ref. 47. Moreover, the Hodge Laplacian cannot be naively extended to the general hypergraph setting.

Thus, we define a new family of Laplacian matrices that can describe a plethora of higher-order relations and maintain a form analogous to the canonical graph Laplacian. We do this by taking inspiration from the general theory of combinatorial complexes[49] and the hypergraph Laplacian[50].

Formally, we fix the diffusion order as $k \in N$ that describes a diffusion process on the $k$-hyperedges of the hypergraph. We then need to decide how the process occurs, specifically how hyperedges are 'connected' so that information can flow between them. The most natural approach is to extract this information from the structure of $\Delta$ by using a notion of adjacency among hyperedges. In general, two $k$-hyperedges $\sigma$ and $\eta$ can be adjacent in two ways:

- $\sigma, \eta$ are $m$-adjacent from above, when there is an $m$-hyperedge $\xi$, with $m > k$, containing both of them: $\xi \supseteq \sigma \cup \eta$;
- $\sigma, \eta$ are $m$-adjacent from below, when they share a common $m$-face $\xi \subseteq \sigma \cap \eta$.

Combining these two definitions, we can define the adjacency number $a_{(k,m)}$ of a pair of distinct $k$-hyperedges $\sigma \neq \eta$ as

$$a_{(k,m)}(\sigma,\eta) = \begin{cases} |\{\lambda \in \Delta_m : \lambda \subseteq \sigma \cap \eta\}| \text{ if } m < k \\ |\{\lambda \in \Delta_m : \sigma \cup \eta \subseteq \lambda\}| \text{ if } m > k \\ 0 \text{ if } m = k \end{cases}, \tag{5}$$

so that we may consider $\eta$ and $\sigma$ to be $m$ adjacent $\sigma \approx_m \eta$ when $a_{(k,m)}(\sigma, \eta) > 0$. Intuitively, $a_{(k,m)}(\sigma, \eta)$ counts the number of $m$-hyperedges connecting $\sigma$ and $\eta$, allowing us to differentiate between different 'strengths' of adjacency. For example, $a_{(2,0)} = 1$ when two triangles share a single vertex and $a_{(2,0)} = 2$ when they share an edge.

The $(k, m)$-adjacency relations can be formalized into different adjacency matrices, analogous to those defined elsewhere[49,81]. If we index the $k$-hyperedges as $\sigma_1, \ldots, \sigma_{n_k}$, we can define the $(k, m)$-adjacency matrix with diffusion order $k$ and interaction order $m$ as the square $n_k \times n_k$ matrix with elements

$$(\mathbf{A}_{(k,m)})_{ij} = a_{(k,m)}(\sigma_i, \sigma_j). \tag{6}$$

The adjacency matrix $\mathbf{A}_{(1,2)}$, for example, describes how edges (1-hyperedges) are connected through triangles (2-hyperedges), whereas $\mathbf{A}_{(3,0)}$ tells us how tetrahedra (3-hyperedges) are attached to one another through vertices (0-hyperedges). Notice how $(k, m)$-adjacency is equivalent to a weighted version of the $q$ nearness described in other work[82–84] when $\Delta$ is a simplicial complex; we restrict our attention only to $k$-simplices.

The matrix $\mathbf{A}_{(k,m)}$ can be seen as the adjacency matrix of a (weighted) graph, which we call the adjacency graph $G_{(k,m)}$ (Fig. 1a), whose nodes are the $k$-hyperedges and the edges are their adjacency relations given by $m$-hyperedges. In particular, $G_{(0,1)}$ is the graph underlying the higher-order network (that is, $\Delta_0 \cup \Delta_1$) and $G_{(1,0)}$ corresponds to its line graph[51]. To each weighted adjacency matrix, we can associate its (weighted) Laplacian, which we name cross-order Laplacian, through the usual formula

$$\mathbf{L}^{\times}_{(k,m)} = \mathrm{diag}(\deg_{(k,m)}) - \mathbf{A}_{(k,m)}, \tag{7}$$

where $\deg_{(k,m)}$ is the vector containing the higher-order $(k, m)$-degrees, defined as the row sums of $\mathbf{A}_{(k,m)}$. More specifically,

$$\deg_{(k,m)}(\sigma) = \sum_{\eta \in \Delta_k} a_{(k,m)}(\sigma, \eta). \tag{8}$$

The matrix $\mathbf{L}^{\times}_{(k,m)}$, being a weighted graph Laplacian, is symmetric, positive semidefinite and has an eigenvalue of 0 with multiplicity given by the number of connected components of its underlying graph $G_{(k,m)}$. We note that our definition includes two existing families of Laplacians as particular cases: (1) the 'vertex-centric' higher-order Laplacians describing how vertices can exchange information through hyperedges (for example, the generalized Laplacians in ref. 39), as cross-order Laplacians of the form $\mathbf{L}^{\times}_{(0,m)}$; and (2) Hodge-like Laplacians in which diffusion happens between hyperedges through the hyperedges of a directly adjacent order, as $\mathbf{L}^{\times}_{(k,k\pm1)}$. Cross-order Laplacians generalize these two notions, allowing the description of diffusion processes, which—as with Hodge Laplacians—can happen on hyperedges of any order and—as with generalized Laplacians—can 'jump' orders, connecting them with hyperedges of any other order.

## Statistical physics of higher-order diffusion processes

**Cross-order diffusion.** From now on, we will consider the high-order diffusion process on $k$-hyperedges through $m$-hyperedges on a given $\Delta$ and, thus, omit the $(k, m)$ notation. Such a process can be easily written as the linear, first-order ordinary differential equation

$$\dot{x}(\tau) = -\mathbf{L}^{\times} x(\tau), \tag{9}$$

where $x(\tau) \in \mathbb{R}^{n_k}$ is a real scalar function on the $k$-hyperedges. Equation (9) can be solved with the time propagation operator (also called the heat kernel) at time $\tau > 0$:

$$\boldsymbol{\rho}(\tau) = \mathrm{e}^{-\tau \mathbf{L}^{\times}}, \tag{10}$$

so that

$$x(\tau) = \boldsymbol{\rho}(\tau) x(0). \tag{11}$$

Due to linearity, the $j$th column $\boldsymbol{\rho}(\tau)_j$ of $\boldsymbol{\rho}(\tau)$ describes the distribution of information over the $k$-hyperedges at time $\tau$, when the total information is concentrated in a single $k$-hyperedge $\sigma_j$ at time $\tau = 0$.

It turns out that we can derive aggregate measures from the heat kernel, for both standard Laplacian on networks[15] and Hodge Laplacians on simplicial complexes[85]. These can help us probe the characteristic scales of the structure under consideration. First, $\boldsymbol{\rho}$ is normalized to a density matrix[15]

$$\hat{\boldsymbol{\rho}}(\tau) = \frac{e^{-\tau \mathbf{L}^{\times}}}{Z(\tau)}, \tag{12}$$

where $Z(\tau) = \mathrm{Tr}(e^{-\tau \mathbf{L}^{\times}})$ is called the return probability, describing how much of the information has remained 'trapped' and did not diffuse at time $\tau$ (ref. 86). With this density matrix, we can compute the von Neumann entropy associated to the diffusion process as

$$S(\tau) = -\mathrm{Tr}(\hat{\boldsymbol{\rho}}(\tau) \log[\hat{\boldsymbol{\rho}}(\tau)]) \tag{13}$$

and its entropic susceptibility[47]

$$C(\tau) = -\frac{dS}{d\log[\tau]}. \tag{14}$$

The density matrix formalism and the associated von Neumann entropy allow us to describe a network's transport properties by simultaneously considering all the possible diffusion trajectories. Thus, we can directly apply it to the cross-order Laplacian to investigate our hypergraph's $k$th-order properties at different scales. This approach, in the $k = 0$ case, has been fruitfully explored in the literature to extract key information about the network's structural organization[15,52–54].

In particular, the maxima and minima of the entropic susceptibility, associated with times of fast deceleration and acceleration of the diffusion process, were shown, in the case of networks, to correspond to the presence of characteristic scales[10,47]. Most importantly, when $C(\tau)$ is constant over a time range $I = [\tau_{\min}, \tau_{\max}]$, we say that $\Delta$ exhibits informational scale invariance[10] in $I$. Explicitly writing orders $(k, m)$ again, it is known that the (network, $k = 0$ and $m = 1$) entropic susceptibility $C_{(0,1)}(\tau)$ shows a large plateau in the case of grid graphs, Barabási–Albert networks and random trees, all of which are examples of self-similar structures. Interestingly, entropic susceptibility and the associated notion of scale invariance can be related to the concept of spectral dimension[55–57], which intuitively measures the dimensionality 'perceived' by a diffusion process taking place on a manifold or, in our case, a graph (Supplementary Section 2).

Note that when the numbers of $k$- and $m$-hyperedges are small, it is possible that little to no $k$-simplices are adjacent through $m$-simplices, resulting in $(k, m)$-adjacency graphs that are very sparse, disconnected and with a high number of isolated nodes. In these settings, the diffusion paths are short and involve few nodes, leading to results that are, in general, less interesting.

**Measuring scale invariance.** Following the considerations above, we want to quantify whether a higher-order network exhibits scale invariance at order $k$ via order $m$. To do so, we define the SIPs $P_{(k,m)}(\epsilon)$ as the logarithmic lifespan of the longest connected plateau of $C_{(k,m)}(\tau)$ with respect to a given tolerance $\epsilon > 0$ on its 'flatness'. In detail, given a value $y > 0$, we define the set

$$E_{(k,m)}(y;\epsilon) = \{\tau > 0 : |\log[C_{(k,m)}(\tau)] - y| < \epsilon\} \tag{15}$$

that—being the inverse image of the open interval $(y - \epsilon, y + \epsilon)$—will be given by a countable, disjoint union of open intervals $(a_i, b_i)$:

$$E_{(k,m)}(y;\epsilon) = \coprod_i (a_i, b_i). \tag{16}$$

We then define the SIP as

$$P_{(k,m)}(\epsilon) = \max_{y>0} \max_i (\log[b_i] - \log[a_i]). \tag{17}$$

If $P_{(k,m)}(\epsilon)$ is large, we can say that the hypergraph is scale invariant at order $k$ via order $m$, whereas if it is close to 0, then $C_{(k,m)}(\tau)$ is peaked, thereby signalling the presence of characteristic scales. Unless stated otherwise, in the numerical experiments, we fix $\epsilon = 0.2$ and omit $\epsilon$ in the notation: $P_{(k,m)}(\epsilon) \equiv P_{(k,m)}$.

## Cross-order Laplacian renormalization scheme

Here we explain in detail our higher-order renormalization algorithm when applied to a general hypergraph $\Delta$. Let us assume that a hypergraph $\Delta$ has been recognized as scale invariant with respect to $C_{(k,m)}(\tau)$ (equation (14)). To reduce it to a smaller, equivalent one that can still be recognized as scale invariant, we need an algorithmic method that does not depend on the specific hypergraph considered. Similar to the case of networks, we can devise a higher-order Laplacian renormalization scheme based on the relationship between $k$-hyperedges through $m$-hyperedges.

In detail, staying faithful to the real-space Laplacian renormalization scheme, we say that $k$-hyperedges $\sigma$ and $\eta$ are to be merged if the information that has flowed from $\sigma$ to $\eta$ is greater than the information that has stayed in $\sigma$ or $\eta$. In practice, we build an auxiliary matrix $\boldsymbol{\zeta}$ with

$$\boldsymbol{\zeta}_{ij} = \begin{cases} 1 & \text{if } \boldsymbol{\rho}_{ij} \geq \min(\boldsymbol{\rho}_{ii}, \boldsymbol{\rho}_{jj}) \\ 0 & \text{otherwise} \end{cases}, \tag{18}$$

which can be thought as the adjacency matrix of a metagraph $\tilde{G}$ in which the nodes represent the $k$-hyperedges and the edges connect those to be identified with respect to the diffusion process. A partition of the $k$-hyperedges is then obtained simply by taking the connected components of $\tilde{G}$.

As prescribed by step (4) discussed in the main text, we need a coarse grain of $\Delta$, which is informed by the partition obtained in step (3).

When the diffusion takes place between the vertices of a network, the process is fairly simple[10]. It is enough to glue together vertices belonging to the same set into super-vertices, and place a super-edge $\{A, B\}$ between the two super-vertices $A$ and $B$ if there are two vertices $a \in A$ and $b \in B$ in the original network such that $\{a, b\} \in \Delta_1$.

If we want to renormalize the higher-order network with respect to a process taking place on general $k$-hyperedges, more care is needed as it is not clear how to naturally collapse hyperedges belonging to the same set in the partition. It is important to highlight that we cannot just coarse grain the adjacency graph with respect to the partition, as that will result in a smaller graph in which most of the relational information contained in the hypergraph has been lost. To approach this problem, we propose the following method which, taking inspiration from ref. 87, aims to reduce the problem to coarse graining a node.

Let us suppose that each $k$-hyperedge $\eta$ has been assigned a label $\Lambda(\eta)$ specifying which set of the partition it belongs to. First, each $k$-hyperedge passes its label down to its vertices, or, from the other point of view, each vertex inherits the labels from all the $k$-hyperedges to which it belongs. Thus, we associate to each vertex $v$ a set, called signature $\Sigma(v)$:

$$\Sigma(v) = \{\Lambda(\eta) | \eta \in \Delta_k, v \in \eta\}, \tag{19}$$

which contains all the labels inherited. The coarse graining is then performed simply by identifying vertices possessing the same signature and then inducing higher-order hyperedges as explained above. If $\Sigma(v) = \emptyset$, meaning that $i$ is not contained in any $k$-hyperedge, then we give it its own signature, so that it will be left unchanged by the transformation.

In other terms, we think of the signature as a labelling of the vertices of the hypergraph, that is, a surjective map $f: \Delta_0 \to S$ from the set of vertices to the set of signatures $S$. As explained in ref. 88 for simplicial complexes, starting from this labelling, we can build another hypergraph $\Delta'$ whose vertices are the signatures, and whose simplices consist of the maps of the simplices of $\Delta$ through $f$, that is,

$$\Delta' = \{\{f(v_0), \ldots, f(v_k)\} | \{v_0, \ldots, v_k\} \in \Delta_k,\ k \in \mathbb{N}\}. \quad (20)$$

Intuitively, step (3) identifies groups of $k$-hyperedges, which can be seen as generalizations of the spin blocks of Kadanoff's renormalization scheme[12]. In general, however, they will not be identical blocks of nodes, but their shape will reflect the structure of $\Delta$ (Fig. 1b–i). Once the partition of the $k$-hyperedges is obtained, we perform a coarse-graining step to aggregate the hyperedges belonging to the same block.

Notice that we cannot directly coarse grain the adjacency graph $G_{(k,m)}$ as we would loose all the relational information between hyperedges of $\Delta$, which is not contained in $G_{(k,m)}$. We, therefore, take a different approach and use the information in the partition of $k$-hyperedges to coarse grain the vertices of $\Delta$. First, we move the problem to the domain of the vertices by letting each one inherit the labels of all the $k$-hyperedges to which they belong (Fig. 1b(ii)). Afterwards, we glue together the vertices that inherited the same label set and induce hyperedges from the starting $\Delta$ (Fig. 1b(iii)). The overall computational complexity of the algorithm is $O(n_k^2 n_m(k+m) + n_k^3 + \sum_{i1}^{k_{\max}} n_i)$ and is discussed in more detail in Supplementary Section 3. Moreover, a visualization of step (3) is provided in Supplementary Section 4.

Note that the choice of time $\tau^*$ heavily influences the renormalization process. If $\tau$ is too low, no vertices will be merged, whereas if $\tau$ grows large enough, every family of connected $k$-hyperedges in $G_{(k,m)}$ will get the same label.

## Data availability

The data supporting the results presented in this Article are available via GitHub at https://github.com/nplresearch/higher_order_LRG.

## Code availability

The code supporting the results presented in this Article are available via GitHub at https://github.com/nplresearch/higher_order_LRG.

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## Author contributions

All authors designed the research and wrote the paper. M.N. and M.M. performed the computational simulations and data analysis.

## Competing interests

The authors declare no competing interests.

## Additional information

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